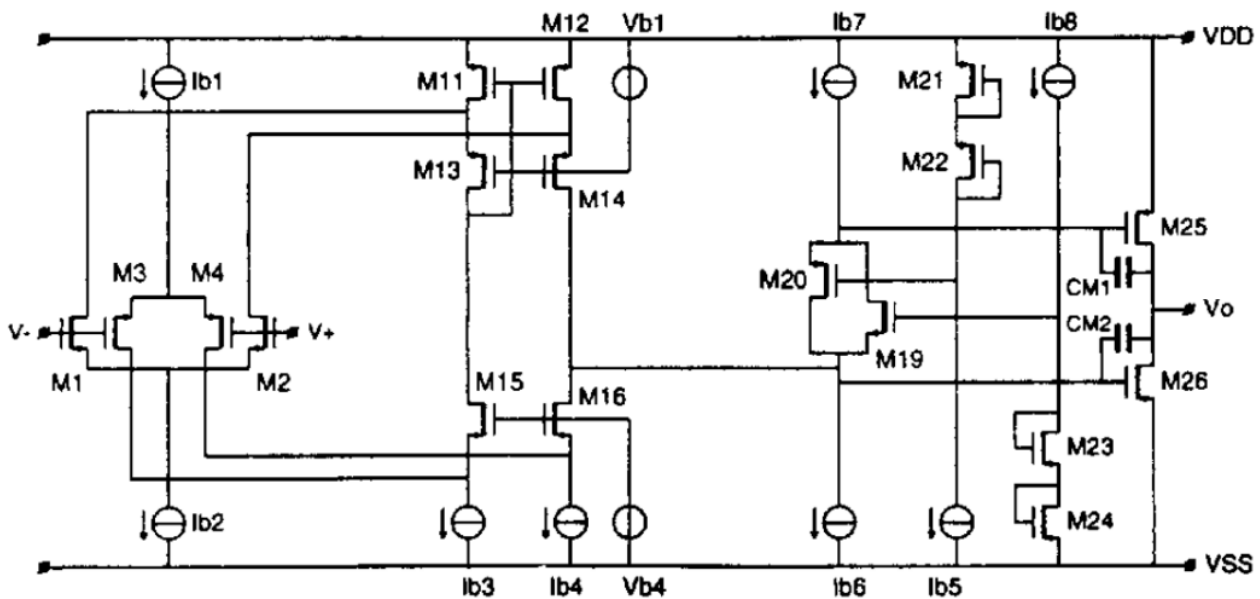


Analog IC Design
Design Challenge 02**Rail-to-Rail Input Translinear-Loop Class-AB output Op-Amp****Intended Learning Objectives**

In this design challenge you will:

- Learn how to design a rail-to-rail input op-amp.
- Learn how to design a Monticelli translinear loop biasing scheme.
- Learn how to design a class-AB output stage.

Design Specs

For the shown op-amp, assume a practical transistor-level implementation for all current sources (i.e., do not assume ideal current sources). For example, I_{b3} can be implemented as a simple current mirror: M_{Ib3} .

Assume:

- $I_{b1} = I_{b2} = I_{b5} = I_{b8} = 10\mu A$
- For the folded-cascode the common-source and the common-gate stages have the same bias current.
- $I_{b7} = 40\mu A$
- Bias current of output stage = $200\mu A$
- Pick reasonable L and g_m/I_D for each transistor

- Use a single ideal current seed, build any bias circuitry other than that (including circuits to generate Vb1 and Vb4).
- Use an ideal voltage source to generate a mid-rail reference voltage $VMID = VDD/2$.

Technology	65nm
Supply voltage	2.5V (use IO devices)
Corners	Temp: 27 Process: TT
RL	10 kOhm to VMID
CL	20 pF
AOL	> 60 dB
PM	> 60 deg
GBW	> 10 MHz

TB1 - Unity-gain buffer (beta = 1) - primary bench

Terminal / component	Connect to
V+ (non-inverting input)	VIN source: vsource dc=1.25 plus the AC / step / sine stimulus
V- (inverting input)	Straight wire from VO, with an iprobe in series for the stb analysis
VO (output)	The feedback wire, RL, and CL
RL far terminal	VMID (1.25 V) - not ground
CL far terminal	GND (VSS)
VIN source return	GND (VSS)

Worst case for phase margin, and the only configuration where VIN, VICM and VO are the same voltage.

TB2 - Non-inverting, G = +2 (beta = 0.5)

Terminal / component	Connect to
V+ (non-inverting input)	VIN source: vsource dc=1.25 plus stimulus
V- (inverting input)	Junction of R1 and R2
R2 = 20 kOhm	From VO to the R1/R2 junction (feedback)
R1 = 20 kOhm	From the junction to VMID - not ground
VO (output)	R2, RL, CL
RL far terminal	VMID (1.25 V)
CL far terminal	GND (VSS)

$VO = VMID + 2 \cdot (VIN - VMID)$.

TB3 - Inverting, G = -1

Terminal / component	Connect to
V+ (non-inverting input)	VMID (1.25 V) directly
V- (inverting input)	Junction of R1 and R2
R1 = 20 kOhm	From the VIN source to the junction
R2 = 20 kOhm	From the junction to VO (feedback)
VO (output)	R2, RL, CL
RL far terminal	VMID (1.25 V)
CL far terminal	GND (VSS)

$VO = VMID - (VIN - VMID)$. The virtual node holds VICM fixed at 1.25 V regardless of signal.

Deliverables

Analysis	Setup	Report	Points
dc + op point	TB1, VICM = VMID	Snapshot showing every device's OP point	2
stb	TB1, VICM = VMID	AOL, fu, PM, GM. Bode plot with the three anchors marked	4
stb, parametric	TB1, VICM swept 0 to 2.5 V, 50 mV step	PM and fu vs VICM	2
dc sweep	TB1, VIN 0.2 to 2.3 V	VO vs VIN ID25 and ID26 vs VIN I _{RL} vs VIN	2
dc sweep	Repeat for RL = 2k	VO vs VIN ID25 and ID26 vs VIN I _{RL} vs VIN	2
tran	TB1, VIN +/-0.5 V step about VMID.	VIN and VO vs time (overlaid) SR+/-	4
tran	TB2 and TB3, VIN sine at 10 kHz. Choose the input amplitude such that output swing is at its maximum value	VIN and VO vs time (overlaid)	4

Compare all deliverables to results expected from hand analysis.

Acknowledgements

Thanks to all who contributed to these labs. If you find any errors or have suggestions concerning these labs, contact Hesham.omran@eng.asu.edu.eg.